

The Duality Calculus: Exact Laws of Fabrication and Blindness for Arithmetic Observers

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Abstract

We develop a two-axis calculus for linear observers of finite residue arithmetic. The first axis, *fabrication*, measures an operator’s capacity to manufacture values outside its data and is controlled by admissibility (positivity); the second, *blindness*, measures what an observer cannot see and is controlled by nothing short of structural access. Theorem 0 proves the axes independent: admissibility never raises visibility, exactness beyond degree one forces negative mass, and “admissible” does not mean “correct.” We diagonalize the loss of divisor-supported charts in the conductor basis, compute the exact Pareto frontier between fabrication budget and loss as a piecewise-quadratic curve with rational breakpoints, and report two laws standing at four exact levels each, including an out-of-sample certification at $P = 30030$: the contact set of the admissible optimum is exactly the even- ω classes, and exact positivity costs precisely the factor $2 - 2/P$. On the multiplicative side we prove a channel dimming law with explicit error bound, an exact price polynomial $Q(x) = \prod_p (\mu_p^2 + x\sigma_p^2)$ for joint observation of parity, and maximal entanglement: visibility 1 requires full joint order. A restoration dichotomy shows typed priors lawfully recover what no in-channel processing can, with machine-certified uniqueness. All computational claims are reproduced by self-contained, dependency-free suites with abort-gated positive and negative controls; every suite has been independently reproduced bit-identically on second hardware. Eight self-corrections, including one conjecture retracted by its own pre-registered out-of-sample test, are published alongside the results. This document is a round-one adversarial-referee target; completeness is not claimed.

1 Introduction: the split

A persistent conflation in operator-based number theory, and in this program’s own earlier record, reads admissibility (nonnegativity, unit mass, bounded envelope) as a warrant of correctness. The present paper is built on the refutation of that reading. There are two independent ways a linear chart can fail: it can *lie* (fabricate values outside the convex hull of its data) and it can *be blind* (fail to see variance its access does not carry). Admissibility cures the first exactly and the second not at all. The calculus below types every observer by both axes, proves the exact trade-off laws between them, prices the escape routes, and audits the historical record under the new typing. Throughout, “admissible” carries the axis tag $[A_1]$ (anti-fabrication only) unless a separate soundness certificate is attached.

2 Setting and axioms

Fix the level k and the primorial $P = P_k = \prod_{i \leq k} p_i$. The substrate is $\mathbb{Z}/P$ with its CRT prime coordinates; the central residual is $\varepsilon_w(N) = \sum_{d|P} w(d)\{N/d\}$ for divisor weights w , the Legendre case being $w = \mu$. A *chart* is a unit-mass linear readout.

F-axioms (fabrication control). F1: unit mass. F2: positivity. F3: bounded envelope constrained away from the identity (genuine aggregation). The *fabrication budget* $\delta(K)$ is the total negative mass; *admissible* means $\delta = 0$.

R-axioms (resolution). R1: a question class is typed by its conductor or channel support. R2: the *visibility* Frac_Q of an observer is the supremum, over estimators built from its data, of explained variance of the target. R3: visibility is monotone in access.

J-axioms (joint observables). J1: channels with explicit laws (multiplicatively, the valuations v_p , independent in density with geometric law). J2: an observer's *grade* is the maximal number of channels any readout couples (the Hoeffding/ANOVA order [4, 5, 6]). J3: object-side typed priors are a lawful channel. J4: every grade increment has an exact positive price (Theorem 8).

3 Theorem 0: independence of the axes

Theorem 1 (Independence). *For unit-mass linear charts constrained away from the identity: (i) No-Lie: $\delta(K) = 0$ implies the maximum principle: outputs lie in the convex hull of the data [10, 11]. (ii) No-Cure: visibility is access-determined. Any function of S -marginal data — in particular any admissible smoothing of it — has explained variance at most $\text{Frac}(S)$, with equality attained by the conditional expectation. (iii) No-Free-Exactness: any chart exact on polynomials of degree ≤ 2 has $\delta(K) > 0$. Consequently δ and Frac are logically independent typings, and the diagonal ($\delta = 0$ and exact beyond degree one) is empty: every such chart loses or lies.*

Proof. (i) is the classical positive-kernel maximum principle. (ii): a smoothing applied to marginal data is itself a function of the marginal data; the L^2 -optimal such function is the conditional expectation, whose explained variance is the spectral projection mass onto accessible conductors (Theorem 2). (iii): a unit-mass positive kernel that is not a point mass has positive second moment $m_2(K)$, so it maps x^2 to $x^2 + m_2 \neq x^2$; exactness forces $m_2 = 0$, excluded by F3; hence exactness requires negative mass. $\square$

Machine witnesses: a Fejér kernel on a step stays in $[0.0000, 1.0000]$ while a sharp kernel with $\delta = 0.509$ emits -0.074 and 1.074 ; two hundred marginal estimators including a Fejér-smoothed profile all respect the ceiling, the best matching it to 10^{-9} [19].

Theorem 2 (Support Determinism). *For the Legendre residual at level k and an observer with prime-coordinate support S ,*

$$\text{Frac}(S) = \frac{2^{|S|} \prod_{p \notin S} (1 - 1/p) - \varphi(P)/P}{2^k - \varphi(P)/P},$$

attained by $\mathbb{E}[\varepsilon | S]$. Verified as exact rational identities for all fifteen nonempty S at $P = 210$ [19]. Machinery: the Hoeffding/Sobol' decomposition [4, 5, 6]; the closed form for this object was not located in the literature search of the campaign's credit gate.

4 The divisor transform

Theorem 3 (Diagonalization). *Let $\theta_w(n) = \sum_{d \in D, d | \gcd(n, P)} w_d$ with $w_1 = 1$ and support budget $D = \{d \mid P : \omega(d) \leq r\}$, and let $L(w) = P^{-1} \sum_n (\theta_w(n) - \mathbf{1}_{\gcd(n, P)=1})^2$. In the conductor basis of*

P -even functions (Ramanujan sums c_c , orthogonal with norm $\varphi(c)$ [3, 2]),

$$L(w) = \sum_{c|P} \varphi(c) \left(W(c) - \frac{\varphi(P)}{P} \frac{\mu(c)}{\varphi(c)} \right)^2, \quad W(c) = \sum_{d \in D, c|d} \frac{w_d}{d}.$$

The objective is diagonal in W ; the support budget is a coordinate subspace in w ; fabrication ($\theta_w \geq -\delta$) is a polyhedral cone in the gcd-class coordinates. The three systems are Möbius-conjugate. At full support the optimum is $w = \mu$ exactly, with $L = 0$ and $\theta = \mathbf{1} \geq 0$: the tension exists only under truncation.

The identity is machine-verified as exact three-way agreement (conductor, class, brute force), with a corrupted-target control failing as required [20].

Theorem 4 (Parseval; exact variance law). $\text{Var}_{\text{period}}(\varepsilon_w) = \sum_{c|P, c>1} W(c)^2 \varphi(c) \psi(c) / 12$, ψ the Dedekind psi, via $\sum_{j \in (\mathbb{Z}/c)^\times} (4 \sin^2(\pi j/c))^{-1} = \varphi(c) \psi(c) / 12$. The instance $w = \mu$ gives $\text{Var} = (\varphi/P)(2^k - \varphi/P) / 12$, verified as exact rational equality for $k \leq 5$ [19]. Adjacency: this instance sits inside reduced-residue second-moment theory [7] and the autocorrelation literature of μ -weighted fractional parts [8, 9]; it is claimed as consolidation with credit, not as novelty.

5 The frontier

Write $L^*(\delta)$ for the minimum of L subject to $\theta_w \geq -\delta$ pointwise.

Theorem 5 (Non-degenerate frontier). *At every truncated support tested, $L^*(0) > L^*(\infty)$ strictly; positivity genuinely costs accuracy. Exact endpoints (KKT-certified rationals): see Table 1. The frontier is piecewise quadratic in δ with exact rational breakpoints, computed by parametric continuation [12] and certified to terminate at $L^*(\infty)$; e.g. at $(k, r) = (3, 2)$ the breakpoints are $2/21, 4/25, 8/29$, the last equal to the maximal fabrication of the unconstrained optimum [20].*

(k, r)	$L^*(\infty)$	$L^*(0)$	(k, r) continued
(3, 1)	16/165	8/45	(5, 1): 1724/13937 $\rightarrow$ 164/693
(3, 2)	4/435	4/225	(5, 2): 8/385 $\rightarrow$ 77351483/2203440717
(4, 1)	4/35	68/315	(5, 3): 64/29337 $\rightarrow$ 10119340094/1826795419295
(4, 2)	4/245	95037/3253180	(5, 4): 16/177793 $\rightarrow$ 16/88935
(4, 3)	8/7315	8/3675	(6, 5): 192/30059029 $\rightarrow$ 64/5010005

Table 1: Exact frontier endpoints; all entries KKT-certified.

Conjecture 1 (Active-Set Law; certified $k = 3, 4, 5, 6$). *At one-level truncation $r = k - 1$ the admissible optimum's contact set is exactly the even- ω classes $e > 1$ (sizes $2^{k-1} - 1$); equivalently the optimal admissible chart is supported on the odd- ω classes together with class 1. The $k = 6$ case ($P = 30030$, 63 variables) was certified out of sample by a single exact KKT solve on the predicted set.*

Conjecture 2 (Factor-Two Law; exact at $k = 3, 4, 5, 6$). *At $r = k - 1$: $L^*(0) = (2 - \frac{2}{P_k}) L^*(\infty)$ exactly. Exact positivity costs precisely the factor two, minus $2/P$.*

Remark 1 (A retraction, preserved). A three-point closed form for the endpoint values (denominator $p_{k-1}p_k$, exact at $k = 3, 4, 5$) was retracted at the pre-registered $k = 6$ out-of-sample test before publication; the true values are $L^*(0) = 2^6/(5 \cdot 1001^2)$ and $L^*(\infty) = 192/((P - 1) \cdot 1001)$ at $k = 6$. The mismatch is preserved in the law suite as a passing control so the retraction cannot be silently undone; the general value law is open, with $k = 7$ pre-registered.

6 The blindness laws

Theorem 6 (Channel Dimming). *For every prime p and $N \geq 1$, $S_p(N) := \sum_{n \leq N} (-1)^{v_p(n)}$ satisfies $|S_p(N) - N \frac{p-1}{p+1}| \leq 2(\log_p N + 1)$.*

Proof. Count by valuation: $S_p(N) = \sum_{j \geq 0} (-1)^j (\lfloor N/p^j \rfloor - \lfloor N/p^{j+1} \rfloor) = N - 2 \sum_{j \geq 1} (-1)^{j-1} \lfloor N/p^j \rfloor$; replace floors by exact ratios, sum the geometric series to $N \frac{p-1}{p+1}$, and bound each of the at most $\log_p N + 1$ floor errors by 1, doubled by the coefficient. $\square$

Machine: errors 0.67–2.00 at $N = 10^6$ for $p \leq 11$ against bounds 13–42; a corrupted factor $(p-1)/p$ fails by 1.7×10^5 [21]. The per-channel mean is a classical Euler-factor computation; the typed transmission framing is the present program’s.

Theorem 7 (Truncated-Parity Blindness). *Let $\lambda_z(n) = (-1)^{\sum_{p \leq z} v_p(n)}$ and let an observer hold the channels S among the primes $\leq z$. Then the correlation of λ_z with the best S -predictor equals $\prod_{p \leq z, p \notin S} \frac{p-1}{p+1}$ up to the Theorem 6 error, the optimal predictor being the channel sign-product, and any function of the S -channel data — including any admissible smoothing — has visibility at most the square of that product (admissibility-independence). For the full Liouville function, the unseen-channel product over all primes diverges to zero (elementary; [17]); the finite- N rate is of prime-number-theorem strength [18] and is cited, not claimed.*

Machine: all eight subsets at $z = 7$ match the exact products (e.g. $S = \{2\}$: measured +0.24992 vs 1/4); two hundred channel estimators including a Fejér smoothing respect the ceiling [21]. The bridge identification $\lambda = (-1)^\Omega$ reads parity as the count of multiplicative gate crossings; parity is thereby a maximally joint observable, which is why every marginal method class meets the same wall.

7 Lawful exceedance

Theorem 8 (Exact price of joint order). *For z -smooth parity with channel set of size K , put $\mu_p = \frac{p-1}{p+1}$, $\sigma_p^2 = \frac{4p}{(p+1)^2}$, and define the price polynomial $Q(x) = \prod_{p \leq z} (\mu_p^2 + x \sigma_p^2) = \sum_{j=0}^K c_j x^j$. Then, with exact rational coefficients: (i) $Q(1) = 1$; (ii) the visibility purchasable at grade m is exactly $V(m) = (c_1 + \dots + c_m)/(1 - c_0)$, every increment strictly positive; (iii) maximal entanglement: $c_K = \prod_p \sigma_p^2 > 0$, so visibility 1 requires grade K ; (iv) an observer joint within S and blind outside has visibility exactly $(\prod_{p \notin S} \mu_p^2 - c_0)/(1 - c_0)$.*

Proof. Write $\lambda_z = \prod_p X_p$, $X_p = (-1)^{v_p}$, channels independent in density with $\mathbb{E}X_p = \mu_p$. The Hoeffding decomposition of a product of independent variables is $\prod_p (\mu_p + (X_p - \mu_p))$ expanded over subsets; strata are orthogonal; the variance of the T -stratum is $\prod_T \sigma_p^2 \prod_{T^c} \mu_p^2$, and the generating function over $|T|$ is $Q(x)$. All four claims are bookkeeping on the coefficients. $\square$

At $z = 7$: $c_{0..4} = \frac{1}{144}, \frac{469}{5184}, \frac{851}{2592}, \frac{2137}{5184}, \frac{35}{216}$ and $V(1..4) = 0.0911, 0.4217, 0.8368, 1$ exactly: separable observers are blind to 90.9% of parity; the wall is a price list [23]. Clause (iv) gives the tight $S = \{2, 3\}$ ceiling $35/143 = 0.24476$, sharpening the 0.25 bound stated at Theorem 7’s suite and bracketing the 0.24415 it measured — a retro-explanation of an instrument reading recorded before the theorem existed.

8 The restoration dichotomy

Theorem 9 (Restoration Dichotomy). *(I) No processing of a lossy readout exceeds the visibility ceiling of its access (Theorem 1). (II) When the unknown object carries a type (here: s -sparse divisor support), exact reconstruction from a lossy readout is possible and certifiable, and when it is not, the failure is certifiable instead.*

Witness at $P = 210$, readout = eight running totals $T(x) = \sum_{n \leq x} f(n) = \sum_d a_d \lfloor x/d \rfloor$ (the line’s native cumulative view): all $\binom{16}{3} = 560$ supports scanned; exactly one exact reconstruction, equal to the hidden object identically; without the prior a second solution is exhibited (data alone cannot restore); a sparsity-5 request returns AMBIGUOUS with witnesses; one corrupted total returns INFEASIBLE; and the first, weaker measurement design is preserved in the suite as a control that must remain ambiguous [22]. Recovery mathematics: compressed sensing [13, 14, 15]; the exact-rational refutation-ledger certificate and the four-verdict instrument are the program’s.

9 Corrections lineage

Published with, not despite, the results: (1) an $O(1)$ equidistribution claim retired (pre-calculus record); (2) the founding split — “barrier = inadmissible artifact” replaced by the two-axis typing; (3) “detection = diagonalization” corrected; (4) the frontier value law retracted at its own pre-registered $k = 6$ test, mismatch preserved as a control; (5) the first restoration design failed its own gate and was preserved as a control; (6) the Theorem 7 suite’s 0.25 ceiling tightened to 35/143 by Theorem 8(iv); (7) the program-wide audit re-typed all comparative accuracy gates as instance-level measurements, never as “admissibility implies accuracy,” with an axis-tag rule made permanent; (8) one suite superseded for violating the single-file standard, supersession recorded.

10 Verification record

Every computational claim is reproduced by self-contained, dependency-free Python suites with abort-gated positive and negative controls; each prints a deterministic result manifest. All suites have been independently reproduced bit-identically on second hardware (five cross-machine certifications). SHA-256 digests:

suite	digest (first/second half)
dc2_verification.py	94b30cbb39dcd1abaed3b35bbdd8d399 6022ce939633076bc4739d1678e492d4
dc3_frontier.py	4ed148f5148a778f0f8e1288e025b3cd 0251104c9a8e857c8d49f431d11d6dbe
dc3_law.py (v2)	025d4ecfff76754e93fadd1e77e43a38 8c6f580035f9c04ac6ec1ed71223110a
dc4_blindness.py	9a074a1264e115bac131753a318a6c86 84d858cb6363ef5475930cfed8de65e6
dc5_restoration_witness.py	06ca334e60e4b48ca433d300bc0d573d c65777de49bea19235554f61620fde9a
dc5_exceedance.py	d7b0d4879fac699d307ff3acfebd08eb 8b88aaeceda18dab04bfad8e15744779

This paper is the round-one target of the program’s two-round adversarial referee protocol; no completeness claim is made before both rounds and the artifact audit pass.

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